AI ASSISTED CODING
ASSIGNMENT - 15.1
ROLL NO. 2303A510C8

Task 1 – Student Records API

Task:

Use AI to build a RESTful API for managing student records.

Instructions:

- **Endpoints required:**
  - **GET /students → List all students**
  - **POST /students → Add a new student**
  - **PUT /students/{id} → Update student details**
  - **DELETE /students/{id} → Delete a student record**
- **Use an in-memory data structure (list or dictionary) to store records.**
- **Ensure API responses are in JSON format.**

Expected Output:

- **Working API with CRUD functionality for student records**

Prompt:

Create a simple REST API in Python using Flask for managing student records.

The API should have these endpoints:

- **GET /students to get all students**
- **POST /students to add a student**
- **PUT /students/<id> to update a student**
- **DELETE /students/<id> to delete a student**

Use a dictionary or list to store data in memory (no database).

Each student should have id, name, age, and course.
ID should be generated automatically.

Make sure:
- All responses are in JSON
- Handle errors like invalid id or missing fields
- Use proper status codes (200, 201, 404, etc.)

Keep the code simple and readable with basic comments (not too fancy).

Also include:
- A small description of the API
- Example requests (like curl or Postman)

The code should run directly without extra setup.

Code:

```
from flask import Flask, jsonify, request

app = Flask(__name__)

# In-memory storage for student records
students = []
next_id = 1


def find_student(student_id):
    for student in students:
        if student["id"] == student_id:
```

```
            return student
    return None


def validate_student_payload(data):
    required_fields = ["name", "age", "course"]
    missing_fields = [field for field in required_fields if field not in data]
    if missing_fields:
        return False, f"Missing fields: {', '.join(missing_fields)}"

    if not isinstance(data["name"], str) or not data["name"].strip():
        return False, "Field 'name' must be non-empty"

    if not isinstance(data["age"], int) or data["age"] <= 0:
        return False, "Field 'age' must be positive integer"

    if not isinstance(data["course"], str) or not data["course"].strip():
        return False, "Field 'course' must be non-empty"

    return True, None


@app.route("/")
def home():
    return jsonify({"message": "Student API is running"}), 200


@app.route("/students", methods=["GET"])
def get_students():
    return jsonify({"students": students}), 200


@app.route("/students", methods=["POST"])
def add_student():
    global next_id

    if not request.is_json:
        return jsonify({"error": "Request must be JSON"}), 400

    data = request.get_json()
    is_valid, error = validate_student_payload(data)
    if not is_valid:
```

```
        return jsonify({"error": error}), 400

    new_student = {
        "id": next_id,
        "name": data["name"],
        "age": data["age"],
        "course": data["course"],
    }

    students.append(new_student)
    next_id += 1

    return jsonify({"message": "Student added", "student": new_student}), 201


@app.route("/students/<int:student_id>", methods=["PUT"])
def update_student(student_id):
    student = find_student(student_id)
    if student is None:
        return jsonify({"error": "Student not found"}), 404

    data = request.get_json()
    is_valid, error = validate_student_payload(data)
    if not is_valid:
        return jsonify({"error": error}), 400

    student["name"] = data["name"]
    student["age"] = data["age"]
    student["course"] = data["course"]

    return jsonify({"message": "Student updated", "student": student}), 200


@app.route("/students/<int:student_id>", methods=["DELETE"])
def delete_student(student_id):
    student = find_student(student_id)
    if student is None:
        return jsonify({"error": "Student not found"}), 404

    students.remove(student)
    return jsonify({"message": "Student deleted"}), 200
```

```python
# ---------------- TEST CASES ----------------
if __name__ == "__main__":
   with app.test_client() as client:
       print("\n--- TESTING API ---")

       # 1. GET empty list
       res = client.get("/students")
       print("GET /students ->", res.json)

       # 2. ADD student
       res = client.post("/students", json={"name": "Rohit", "age": 20, "course":
"CSE"})
       print("POST /students ->", res.json)

       # 3. GET after adding
       res = client.get("/students")
       print("GET /students ->", res.json)

       # 4. UPDATE student
       res = client.put("/students/1", json={"name": "Rohit Kumar", "age": 21,
"course": "IT"})
       print("PUT /students/1 ->", res.json)

       # 5. DELETE student
       res = client.delete("/students/1")
       print("DELETE /students/1 ->", res.json)

       # 6. GET after delete
       res = client.get("/students")
       print("GET /students ->", res.json)

   app.run(debug=True)
```

Output:

```
--- TESTING API ---
GET /students -> {'students': []}
POST /students -> {'message': 'Student
```

```
added', 'student': {'age': 20,
'course': 'CSE', 'id': 1, 'name':
'Rohit'}}
GET /students -> {'students': [{'age':
20, 'course': 'CSE', 'id': 1, 'name':
'Rohit'}]}
PUT /students/1 -> {'message': 'Student
updated', 'student': {'age': 21,
'course': 'IT', 'id': 1, 'name': 'Rohit
Kumar'}}
DELETE /students/1 -> {'message':
'Student deleted'}
GET /students -> {'students': []}
 * Debugger is active!
 * Debugger PIN: 142-074-111
127.0.0.1 - - [23/Mar/2026 10:04:08]
"GET / HTTP/1.1" 200 -
```

Task 2 – Library Book Management API

Task:

Develop a RESTful API to handle library books.

Instructions:

- **Endpoints required:**
  - **GET /books → Retrieve all books**
  - **POST /books → Add a new book**
  - **GET /books/{id} → Get details of a specific book**
  - **PATCH /books/{id} → Update partial book details (e.g., availability)**
  - **DELETE /books/{id} → Remove a book**
- **Implement error handling for invalid requests.**

Expected Output:

- **Functional API with CRUD + partial updates.**

Prompt:Create an API for managing employees and their salaries.

Instructions:

- **Endpoints required:**
  - **GET /employees → List all employees**
  - **POST /employees → Add a new employee with salary details**
  - **PUT /employees/{id}/salary → Update salary of an employee**
  - **DELETE /employees/{id} → Remove employee from system**
- **Use AI to:**
  - **Suggest data model structure.**

o Add comments/docstrings for all endpoints.
Expected Output:
• API supporting salary management with clear documentation.
Prompt:
Create a simple REST API in Python using Flask to manage library books.

The API should support the following endpoints:
- GET /books → get all books
- POST /books → add a new book
- GET /books/<id> → get a specific book
- PATCH /books/<id> → update partial details (like availability)
- DELETE /books/<id> → delete a book

Use an in-memory list or dictionary to store data.

Each book should have:
id, title, author, available (boolean)

Requirements:
- ID should be auto-generated
- All responses must be in JSON
- Handle errors like invalid id or missing fields
- Use proper status codes (200, 201, 404, 400)
- Keep code simple with basic comments (not too complex)

Also include:
- Short description of API
- Example test cases or outputs

The code should run directly.

Code:

```
from flask import Flask, jsonify, request

app = Flask(__name__)

books = []
next_id = 1


def find_book(book_id):
    for book in books:
        if book["id"] == book_id:
            return book
    return None


@app.route("/")
def home():
    return jsonify({"message": "Library API running"})


# GET all books
@app.route("/books", methods=["GET"])
def get_books():
    return jsonify({"books": books}), 200


# ADD book
@app.route("/books", methods=["POST"])
def add_book():
    global next_id
```

```python
    data = request.get_json()

    if not data or "title" not in data or "author" not in data:
        return jsonify({"error": "Missing title or author"}), 400

    new_book = {
        "id": next_id,
        "title": data["title"],
        "author": data["author"],
        "available": data.get("available", True)
    }

    books.append(new_book)
    next_id += 1

    return jsonify({"message": "Book added", "book": new_book}), 201


# GET single book
@app.route("/books/<int:book_id>", methods=["GET"])
def get_book(book_id):
    book = find_book(book_id)
    if not book:
        return jsonify({"error": "Book not found"}), 404

    return jsonify(book), 200


# PATCH (partial update)
@app.route("/books/<int:book_id>", methods=["PATCH"])
def update_book(book_id):
    book = find_book(book_id)
    if not book:
        return jsonify({"error": "Book not found"}), 404

    data = request.get_json()

    if "title" in data:
        book["title"] = data["title"]
    if "author" in data:
        book["author"] = data["author"]
```

```python
    if "available" in data:
        book["available"] = data["available"]

    return jsonify({"message": "Book updated", "book": book}), 200


# DELETE book
@app.route("/books/<int:book_id>", methods=["DELETE"])
def delete_book(book_id):
    book = find_book(book_id)
    if not book:
        return jsonify({"error": "Book not found"}), 404

    books.remove(book)
    return jsonify({"message": "Book deleted"}), 200


# TEST CASES
if __name__ == "__main__":
    with app.test_client() as client:
        print("\n--- TESTING BOOK API ---")

        # 1. GET empty
        print("GET /books ->", client.get("/books").json)

        # 2. ADD book
        print("POST /books ->", client.post("/books", json={
            "title": "Python Basics",
            "author": "John"
        }).json)

        # 3. GET all
        print("GET /books ->", client.get("/books").json)

        # 4. GET one
        print("GET /books/1 ->", client.get("/books/1").json)

        # 5. PATCH update
        print("PATCH /books/1 ->", client.patch("/books/1", json={
            "available": False
        }).json)
```

```python
    # 6. DELETE
    print("DELETE /books/1 ->", client.delete("/books/1").json)

    # 7. GET after delete
    print("GET /books ->", client.get("/books").json)

app.run(debug=True)
```

Output:

```
* Restarting with stat

--- TESTING BOOK API ---
GET /books -> {'books': []}
POST /books -> {'book': {'author': 'John',
'available': True, 'id': 1, 'title': 'Python
Basics'}, 'message': 'Book added'}
GET /books -> {'books': [{'author': 'John',
'available': True, 'id': 1, 'title': 'Python
Basics'}]}
GET /books/1 -> {'author': 'John', 'available':
True, 'id': 1, 'title': 'Python Basics'}
PATCH /books/1 -> {'book': {'author': 'John',
'available': False, 'id': 1, 'title': 'Python
Basics'}, 'message': 'Book updated'}
DELETE /books/1 -> {'message': 'Book deleted'}
GET /books -> {'books': []}
 * Debugger is active!
```

```
* Debugger PIN: 142-074-111
```

Task 3 – Employee Payroll API
Task:
Create an API for managing employees and their salaries.
Instructions:
• Endpoints required:
o GET /employees → List all employees
o POST /employees → Add a new employee with salary details
o PUT /employees/{id}/salary → Update salary of an employee
o DELETE /employees/{id} → Remove employee from system
• Use AI to:
o Suggest data model structure.
o Add comments/docstrings for all endpoints.
Expected Output:
• API supporting salary management with clear documentation.
Prompt:
Create a simple REST API in Python using Flask to manage employees and their salaries.

Endpoints required:
- GET /employees → list all employees
- POST /employees → add a new employee with salary
- PUT /employees/<id>/salary → update salary of an employee
- DELETE /employees/<id> → delete an employee

Use an in-memory list or dictionary.

Each employee should have:
id, name, role, salary

Requirements:
- Auto-generate ID
- Return all responses in JSON
- Handle errors like invalid ID or missing fields
- Use proper status codes
- Keep code simple and readable

Also:
- Suggest a basic data model structure
- Add comments/docstrings for all endpoints
- Include test cases showing API working

The code should run directly without extra setup.

Code:

```
from flask import Flask, jsonify, request


app = Flask(__name__)
```

```python
employees = []
next_id = 1


def find_employee(emp_id):
    """Find employee by ID"""
    for emp in employees:
        if emp["id"] == emp_id:
            return emp
    return None


@app.route("/")
def home():
    return jsonify({"message": "Employee Payroll API running"})


# GET all employees
@app.route("/employees", methods=["GET"])
def get_employees():
    """Return list of all employees"""
    return jsonify({"employees": employees}), 200


# ADD employee
@app.route("/employees", methods=["POST"])
def add_employee():
    """Add a new employee with salary"""
    global next_id

    data = request.get_json()

    if not data or "name" not in data or "role" not in data or "salary" not in data:
        return jsonify({"error": "Missing fields"}), 400

    new_emp = {
        "id": next_id,
        "name": data["name"],
        "role": data["role"],
        "salary": data["salary"]
    }
```

```python
    employees.append(new_emp)
    next_id += 1

    return jsonify({"message": "Employee added", "employee": new_emp}), 201


# UPDATE salary
@app.route("/employees/<int:emp_id>/salary", methods=["PUT"])
def update_salary(emp_id):
    """Update salary of an employee"""
    emp = find_employee(emp_id)
    if not emp:
        return jsonify({"error": "Employee not found"}), 404

    data = request.get_json()

    if not data or "salary" not in data:
        return jsonify({"error": "Salary required"}), 400

    emp["salary"] = data["salary"]

    return jsonify({"message": "Salary updated", "employee": emp}), 200


# DELETE employee
@app.route("/employees/<int:emp_id>", methods=["DELETE"])
def delete_employee(emp_id):
    """Remove employee from system"""
    emp = find_employee(emp_id)
    if not emp:
        return jsonify({"error": "Employee not found"}), 404

    employees.remove(emp)
    return jsonify({"message": "Employee deleted"}), 200


# ---------------- TEST CASES ----------------
if __name__ == "__main__":
    with app.test_client() as client:
        print("\n--- TESTING EMPLOYEE API ---")
```

```python
    # 1. GET empty
    print("GET /employees ->", client.get("/employees").json)

    # 2. ADD employee
    print("POST /employees ->", client.post("/employees", json={
        "name": "Rohit",
        "role": "Developer",
        "salary": 50000
    }).json)

    # 3. GET all
    print("GET /employees ->", client.get("/employees").json)

    # 4. UPDATE salary
    print("PUT /employees/1/salary ->", client.put("/employees/1/salary", json={
        "salary": 60000
    }).json)

    # 5. DELETE employee
    print("DELETE /employees/1 ->", client.delete("/employees/1").json)

    # 6. GET after delete
    print("GET /employees ->", client.get("/employees").json)

app.run(debug=True)
```

Output:

```
GET /employees -> {'employees': []}
POST /employees -> {'employee': {'id': 1,
'name': 'Rohit', 'role': 'Developer', 'salary':
50000}, 'message': 'Employee added'}
GET /employees -> {'employees': [{'id': 1,
'name': 'Rohit', 'role': 'Developer', 'salary':
```

```
50000}]}
PUT /employees/1/salary -> {'employee': {'id':
1, 'name': 'Rohit', 'role': 'Developer',
'salary': 60000}, 'message': 'Salary updated'}
DELETE /employees/1 -> {'message': 'Employee
deleted'}
GET /employees -> {'employees': []}
 * Debugger is active!
 * Debugger PIN: 142-074-111
/usr/local/bin/python3
/Users/srinivas/Desktop/HCP/Assignment-15.1.py
```

Task 4 – Real-Time Application: Online Food Ordering API
Scenario:
Design a simple API for an online food ordering system.
Requirements:
• Endpoints required:
o GET /menu → List available dishes
o POST /order → Place a new order
o GET /order/{id} → Track order status
o PUT /order/{id} → Update an existing order (e.g., change items)
o DELETE /order/{id} → Cancel an order

• AI should generate:
o REST API code
o Suggested improvements (like authentication, pagination)
Expected Output:
• Fully working API simulating a food ordering backend.
 Deliverables (For All Tasks)
1. AI-generated prompts for code and test case generation.
2. At least 3 assert test cases for each task.
3. AI-generated initial code and execution screenshots.
4. Analysis of whether code passes all tests.
5. Improved final version with inline comments and explanation.
6. Compiled report (Word/PDF) with prompts, test cases,

Prompt:
Create a simple REST API in Python using Flask for an online food ordering system.

Endpoints required:
- GET /menu → list all available dishes
- POST /order → place a new order
- GET /order/<id> → track order status
- PUT /order/<id> → update an order (change items)
- DELETE /order/<id> → cancel an order

Use in-memory storage (list or dictionary).

Each order should have:
id, items (list), total_price, status

Menu can be predefined with some dishes and prices.

Requirements:
- Auto-generate order ID
- Calculate total price based on items
- All responses in JSON
- Handle errors (invalid id, wrong items)
- Use proper status codes
- Keep code simple with comments

Also include:
- Short description
- Test cases showing API working
- Some suggested improvements (like authentication, pagination)

The code should run directly.

Code:

```
from flask import Flask, jsonify, request

app = Flask(__name__)

# Predefined menu
menu = {
   "pizza": 200,
   "burger": 100,
```

```
    "pasta": 150
}

orders = []
next_id = 1


def find_order(order_id):
    for order in orders:
        if order["id"] == order_id:
            return order
    return None


def calculate_total(items):
    total = 0
    for item in items:
        if item not in menu:
            return None
        total += menu[item]
    return total


@app.route("/")
def home():
    return jsonify({"message": "Food Ordering API running"})


# GET menu
@app.route("/menu", methods=["GET"])
def get_menu():
    return jsonify({"menu": menu}), 200


# PLACE order
@app.route("/order", methods=["POST"])
def place_order():
    global next_id

    data = request.get_json()

    if not data or "items" not in data:
```

```
        return jsonify({"error": "Items required"}), 400

    total = calculate_total(data["items"])
    if total is None:
        return jsonify({"error": "Invalid item in order"}), 400

    new_order = {
        "id": next_id,
        "items": data["items"],
        "total_price": total,
        "status": "placed"
    }

    orders.append(new_order)
    next_id += 1

    return jsonify({"message": "Order placed", "order": new_order}), 201


# GET order
@app.route("/order/<int:order_id>", methods=["GET"])
def get_order(order_id):
    order = find_order(order_id)
    if not order:
        return jsonify({"error": "Order not found"}), 404

    return jsonify(order), 200


# UPDATE order
@app.route("/order/<int:order_id>", methods=["PUT"])
def update_order(order_id):
    order = find_order(order_id)
    if not order:
        return jsonify({"error": "Order not found"}), 404

    data = request.get_json()

    if not data or "items" not in data:
        return jsonify({"error": "Items required"}), 400

    total = calculate_total(data["items"])
```

```python
    if total is None:
        return jsonify({"error": "Invalid item"}), 400

    order["items"] = data["items"]
    order["total_price"] = total

    return jsonify({"message": "Order updated", "order": order}), 200


# DELETE order
@app.route("/order/<int:order_id>", methods=["DELETE"])
def delete_order(order_id):
    order = find_order(order_id)
    if not order:
        return jsonify({"error": "Order not found"}), 404

    orders.remove(order)
    return jsonify({"message": "Order cancelled"}), 200


# ---------------- TEST CASES ----------------
if __name__ == "__main__":
    with app.test_client() as client:
        print("\n--- TESTING FOOD API ---")

        # 1. GET menu
        print("GET /menu ->", client.get("/menu").json)

        # 2. PLACE order
        print("POST /order ->", client.post("/order", json={
            "items": ["pizza", "burger"]
        }).json)

        # 3. GET order
        print("GET /order/1 ->", client.get("/order/1").json)

        # 4. UPDATE order
        print("PUT /order/1 ->", client.put("/order/1", json={
            "items": ["pasta"]
        }).json)

        # 5. DELETE order
```

```
    print("DELETE /order/1 ->", client.delete("/order/1").json)

    # 6. GET after delete
    print("GET /order/1 ->", client.get("/order/1").json)

app.run(debug=True)
```

Output:

```
--- TESTING FOOD API ---
GET /menu -> {'menu': {'burger': 100, 'pasta':
150, 'pizza': 200}}
POST /order -> {'message': 'Order placed',
'order': {'id': 1, 'items': ['pizza', 'burger'],
'status': 'placed', 'total_price': 300}}
GET /order/1 -> {'id': 1, 'items': ['pizza',
'burger'], 'status': 'placed', 'total_price':
300}
PUT /order/1 -> {'message': 'Order updated',
'order': {'id': 1, 'items': ['pasta'], 'status':
'placed', 'total_price': 150}}
DELETE /order/1 -> {'message': 'Order
cancelled'}
GET /order/1 -> {'error': 'Order not found'}
 * Debugger is active!
 * Debugger PIN: 142-074-111
```